

Text Analytics 635AA

A.Y. 2025/26 Written test September, 04, 2026

Deliver results within 2 h from start time. No teaching material is allowed during the test

Exercise 1 [8 pts].

- (i) Describe the tasks of *sentence splitting* and *tokenization*, explaining the goal of each one and why they are placed at the beginning of the NLP processing pipeline. Explain why neither of the two is a trivial task, giving concrete examples of cases in which a naive implementation produces wrong results. (3 points)
 - (ii) Describe *POS tagging*: what it assigns to each token, and which kind of classes are used. Explain, with an example, how POS tags help to reduce lexical ambiguity. (2 points)
 - (iii) Describe NER: what the task consists of, what it produces as output, and which entity types are typically considered. Discuss two applications of NER and what makes the task hard. (3 points)
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Exercise 2 [8 pts].

- (i) Define what a probabilistic language model is. Write the decomposition of the probability of a word sequence given by the chain rule of probability, and explain why this exact formulation, although always valid, cannot be used in practice to model a language. (3 points)
 - (ii) Describe the unigram model and the independence assumption, and explain how its parameters are estimated from a corpus by MLE. Then state the Markov property, explain how it leads from the unigram model to n-gram models, and write the bigram approximation of the probability of a sequence. (3 points)
 - (iii) Consider a sequence of two adjacent tokens, such as “New York” or “of the”. Explain the difference between calling such a pair a *bigram* and calling it a *collocation*. In particular, explain what additional condition a two-token sequence must satisfy in order to be considered a collocation. (2 points)
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Exercise 3 [8 pts].

- (i) Describe Word2Vec. Explain the prediction task it solves. Describe the architecture: how words enter the network, the role of the weight matrices, and what the hidden layer represents. Then explain the difference between the Skip-gram and the CBOW prediction tasks. (4 points)
 - (ii) Describe how the model is trained: what is kept at the end of training and what is discarded? (2 points)
 - (iii) Discuss the main limitations of Word2Vec embeddings, in particular with respect to out-of-vocabulary words. Explain how the use of subword information addresses some of these limitations. (2 points)
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Exercise 4 [8 pts].

- (i) Explain the purpose of an experimental protocol in the evaluation of a text analytics method. Describe the train and test protocol, explaining the role of the training set and of the test set, and state clearly the fundamental rule concerning the use of test data. Explain why a separate validation (development) set is needed when the parameters of the pipeline must be chosen, and describe how the final model is obtained and evaluated once the best parameters have been identified. (3 points)
- (ii) Describe k-fold cross-validation and leave-one-out validation, discussing for each of them the advantages and disadvantages in terms of computational cost, reproducibility, and statistical relevance of the results. Then explain what *stratification* is and when it is important. (3 points)
- (iii) Explain what parameter optimization by grid search consists of, and why the number of configurations to be tested can easily explode. Describe a strategy that can be used to keep the search feasible. (2 points)